

MODULE-11 : MANAGING PROCESSES

GUIDED BY : VIPUL MISTRI

Q.1 Open your Linux terminal and use the ps command to display all currently running processes for your user. Save the output to a file named my_processes.txt.

ANS :

```
[root@localhost /]# pa >> myprocess.txt
bash: pa: command not found...
[root@localhost /]# ps >> myprocess.txt
[root@localhost /]# cat myprocess.txt
  PID TTY          TIME CMD
 43917 pts/0        00:00:00 bash
 44224 pts/0        00:00:00 ps
[root@localhost /]# ps
  PID TTY          TIME CMD
 43917 pts/0        00:00:00 bash
 44236 pts/0        00:00:00 ps
[root@localhost /]#
```

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Q.2 Start the Spotify client (or any music player app available on your Linux system), then use the top command to find its process ID (PID) and note it down. Hint: Look for the process name in the top list or use pgrep if needed.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal
Aug 21 15:46
ismail@localhost:~$
alias          attrack      chapter
chapter_n     chapter_p    clear
crop          delete      description
enqueue       f                F
faster        fastforward  frame
fullscreen    get_length     get_time
get_title     goto           gotoitem
help          hotkey         info
is_playing    key            license
lock          logout         longhelp
loop          move           next
normal        pause         play
playlist      prev          quit
random        rate          repeat
rewind        save_env      sd
search        seek          set
shutdown      slower        snapshot
sort          stats         status
stop          strack        title
title_n       title_p       vcrop
vdeinterlace  vdeinterlace_mode vlm
voidown       volume        volup
vratio        vtrack        vzoom
zoom

> play
> [00005b5417a6b20] main playlist: playlist is empty
exit
Error in 'exit': libvlc's "exit" is not a command
Unknown command 'exit'. Type 'help' for help.
> ^[[^[[^[[
Error in 'ibvlc's "s not a command
Unknown command 'ype 'help' for help.
> ^Z
[1]+  Stopped                  vlc
ismail@localhost ~/]$ ps
  PID TTY          TIME CMD
 44600 pts/0    00:00:00 bash
 44631 pts/0    00:00:00 vlc
 44705 pts/0    00:00:00 ps
ismail@localhost ~/]$ top
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal
Aug 21 15:47
ismail@localhost:~$ top
top - 15:47:48 up 1 day, 2:49, 2 users, load average: 0.50, 0.21, 0.09
Tasks: 310 total, 2 running, 307 sleeping, 1 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3626.0 total, 683.5 free, 1562.9 used, 1636.4 buff/cache
MiB Swap: 4012.0 total, 3870.2 free, 141.8 used, 2063.0 avail Mem

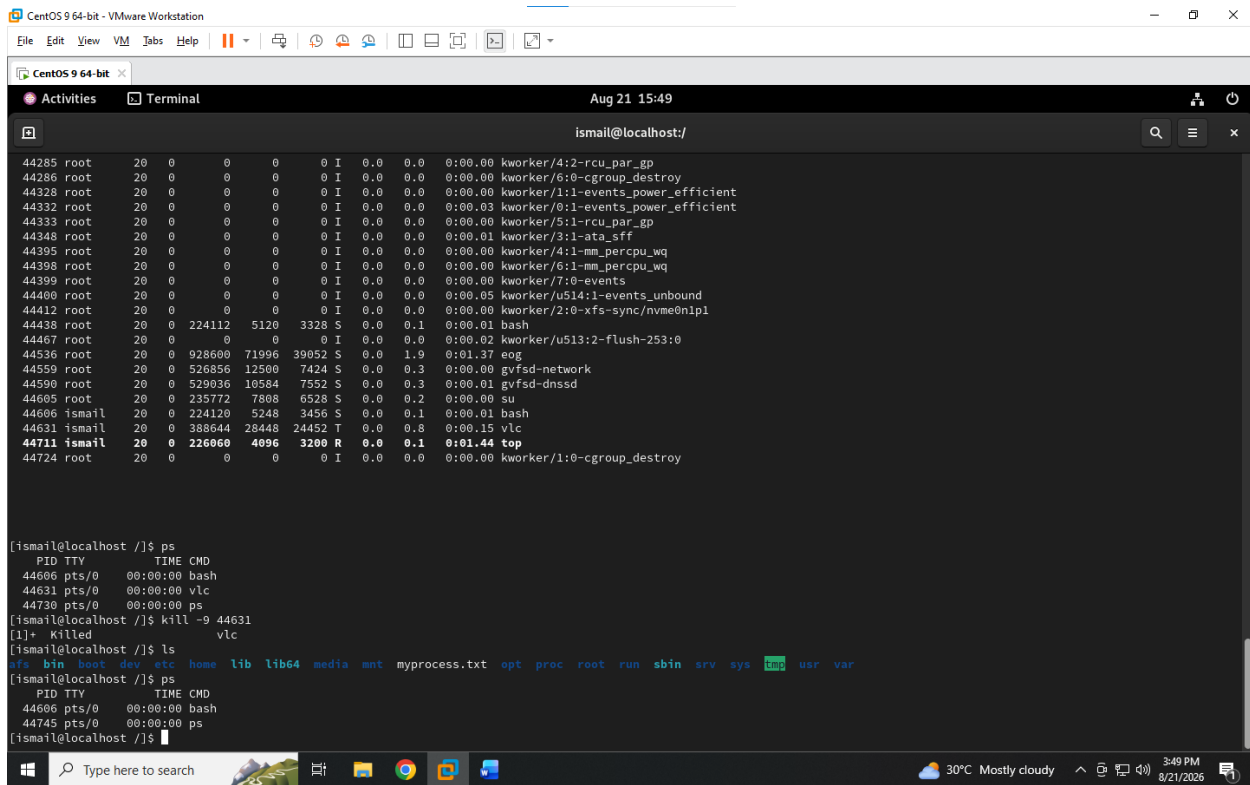
  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 41796 root        20   0     0     0     0  I   0.0   0.0   0:00.06 kworker/u513:3-events_unbound
 41807 root        20   0     0     0     0  I   0.0   0.0   0:00.18 kworker/u514:3-flush-253:0
 43973 root        20   0     0     0     0  I   0.0   0.0   0:00.01 kworker/u514:2-events_unbound
 43974 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/1:2-cgroup_destroy
 43981 root        20   0     0     0     0  I   0.0   0.0   0:00.02 kworker/3:0-ata_sff
 44171 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/2:3-cgwb_release
 44272 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/7:1-events
 44285 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/4:2-rcu_par_gp
 44286 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/6:0-cgroup_destroy
 44328 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/1:1-mm_percpu_wq
 44332 root        20   0     0     0     0  I   0.0   0.0   0:00.02 kworker/0:1-events_power_efficient
 44333 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/5:1-rcu_par_gp
 44348 root        20   0     0     0     0  I   0.0   0.0   0:00.01 kworker/3:1-ata_sff
 44395 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/4:1-mm_percpu_wq
 44398 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/6:1-events
 44399 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/7:0-events
 44400 root        20   0     0     0     0  I   0.0   0.0   0:00.05 kworker/u514:1-events_unbound
 44412 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/2:0-events
 44438 root        20   0 224112 5120 3328 S   0.0   0.1   0:00.01 bash
 44467 root        20   0     0     0     0  I   0.0   0.0   0:00.02 kworker/u513:2-flush-253:0
 44536 root        20   0 928600 71996 39052 S   0.0   1.9   0:01.37 eog
 44559 root        20   0 526856 12500 7424 S   0.0   0.3   0:00.00 gvfsd-network
 44590 root        20   0 529036 10584 7552 S   0.0   0.3   0:00.01 gvfsd-dnssd
 44605 root        20   0 235772 7808 6528 S   0.0   0.2   0:00.00 su
 44606 ismail       20   0 224120 5248 3456 S   0.0   0.1   0:00.01 bash
 44631 ismail       20   0 388644 28448 24452 T   0.0   0.8   0:00.15 vlc
 44711 ismail       20   0 226060 4096 3200 R   0.0   0.1   0:01.37 top
 44724 root        20   0     0     0     0  I   0.0   0.0   0:00.00 kworker/1:0-cgroup_destroy
```

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Q.3 Use the kill command to terminate the process you found in the previous step. Verify that the process has stopped by running ps or top again.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 21 15:49
ismail@localhost:/

44285 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/4:2-rcu_par_gp
44286 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/6:0-cgroup_destroy
44328 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/1:1-events_power_efficient
44332 root 20 0 0 0 0 I 0.0 0.0 0:00.03 kworker/0:1-events_power_efficient
44333 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/5:1-rcu_par_gp
44348 root 20 0 0 0 0 I 0.0 0.0 0:00.01 kworker/3:1-ata_sff
44395 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/4:1-mm_percpu_wq
44398 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/6:1-mm_percpu_wq
44399 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/7:0-events
44400 root 20 0 0 0 0 I 0.0 0.0 0:00.05 kworker/u514:1-events_unbound
44412 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/2:0-xfs-sync/nvme0n1p1
44438 root 20 0 224112 5120 3328 S 0.0 0.1 0:00.01 bash
44467 root 20 0 0 0 0 I 0.0 0.0 0:00.02 kworker/u513:2-flush-253:0
44536 root 20 0 928600 71996 39052 S 0.0 1.9 0:01.37 eog
44559 root 20 0 526856 12500 7424 S 0.0 0.3 0:00.00 gvfsd-network
44590 root 20 0 529036 10584 7552 S 0.0 0.3 0:00.01 gvfsd-dnssd
44605 root 20 0 235772 7808 6528 S 0.0 0.2 0:00.00 su
44606 ismail 20 0 224120 5248 3456 S 0.0 0.1 0:00.01 bash
44631 ismail 20 0 388644 28448 24452 T 0.0 0.8 0:00.15 vlc
44711 ismail 20 0 226060 4096 3200 R 0.0 0.1 0:01.44 top
44724 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/1:0-cgroup_destroy

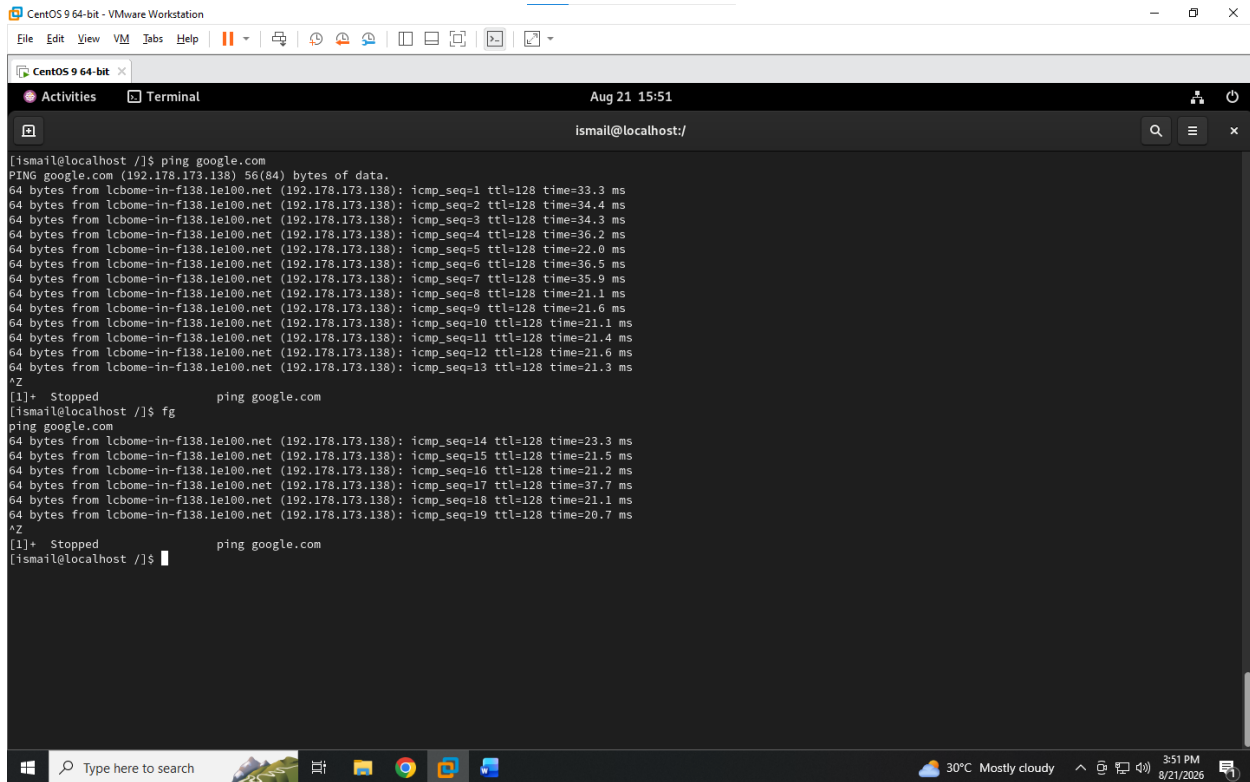
[ismail@localhost ~]$ ps
  PID TTY          TIME CMD
 44606 pts/0    00:00:00 bash
 44631 pts/0    00:00:00 vlc
 44730 pts/0    00:00:00 ps
[ismail@localhost ~]$ kill -9 44631
[1]+  killed                  vlc
[ismail@localhost ~]$ ls
afs bin boot dev etc home lib lib64 media mnt myprocess.txt opt proc root run sbin srv sys tmp usr var
[ismail@localhost ~]$ ps
  PID TTY          TIME CMD
 44606 pts/0    00:00:00 bash
 44745 pts/0    00:00:00 ps
[ismail@localhost ~]$
```

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Q.4 Run a long-running command like ping google.com in the background using &. Then, bring it to the foreground and stop it using Ctrl+Z and the fg command. Hint: Use the jobs command to see background jobs.

ANS :



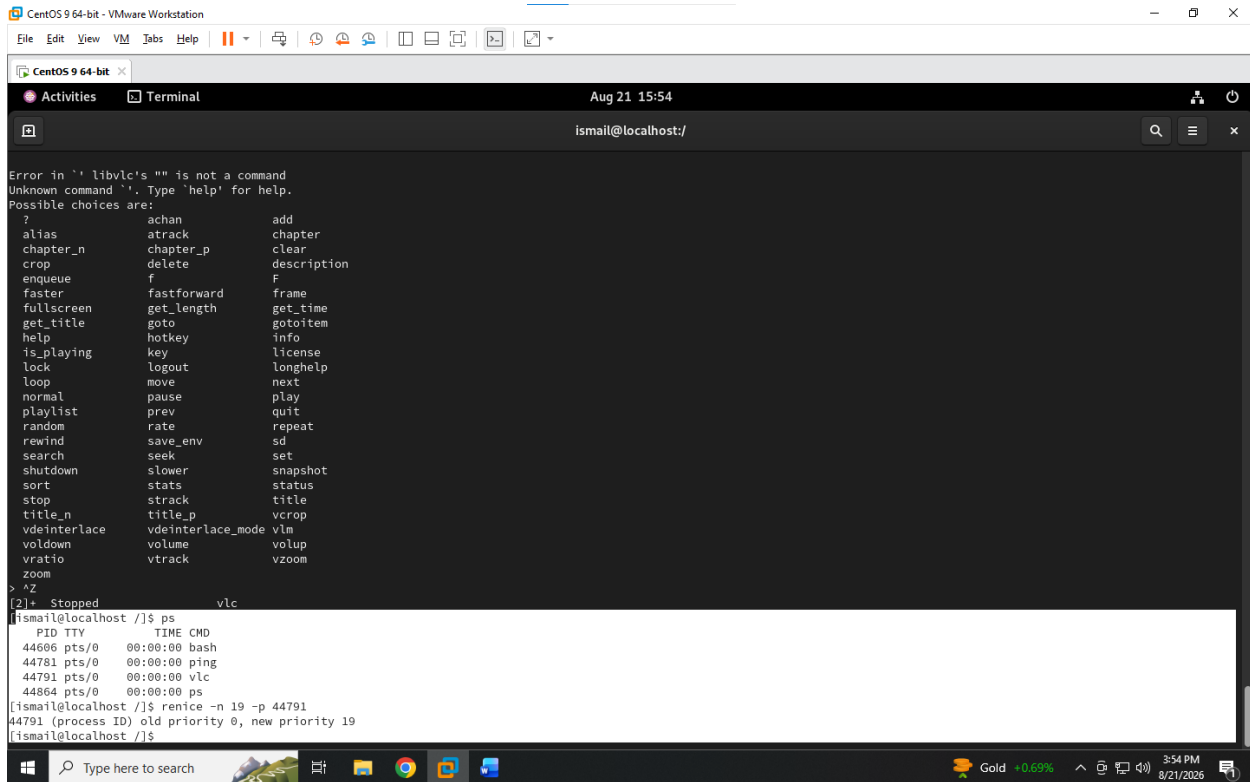
```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Jobs Help
CentOS 9 64-bit
Activities Terminal Aug 21 15:51
ismail@localhost:~$ ping google.com
PING google.com (192.178.173.138) 56(84) bytes of data:
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=1 ttl=128 time=33.3 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=2 ttl=128 time=34.4 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=3 ttl=128 time=34.3 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=4 ttl=128 time=36.2 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=5 ttl=128 time=22.0 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=6 ttl=128 time=36.5 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=7 ttl=128 time=35.9 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=8 ttl=128 time=21.1 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=9 ttl=128 time=21.6 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=10 ttl=128 time=21.1 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=11 ttl=128 time=21.4 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=12 ttl=128 time=21.6 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=13 ttl=128 time=21.3 ms
^Z
[1]+  Stopped                  ping google.com
ismail@localhost:~$ fg
ping google.com
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=14 ttl=128 time=23.3 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=15 ttl=128 time=21.5 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=16 ttl=128 time=21.2 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=17 ttl=128 time=37.7 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=18 ttl=128 time=21.1 ms
64 bytes from lcbome-in-f138.1e100.net (192.178.173.138): icmp_seq=19 ttl=128 time=20.7 ms
^Z
[1]+  Stopped                  ping google.com
ismail@localhost:~$
```

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Q.5 : Suppose you accidentally started a heavy process that is slowing down your system (e.g., a big file copy or a simulation). Use the nice and renice commands to lower its priority and explain in one line how this helps system performance.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal Aug 21 15:54 ismail@localhost:/

Error in `': libvlc's "" is not a command
Unknown command `'. Type 'help' for help.
Possible choices are:
?          achan          add
alias      atrack          chapter
chapter_n  chapter_p         clear
crop       delete         description
enqueue    f                 F
faster     fastforward       frame
fullscreen get_length        get_time
get_title  goto             gotoitem
help       hotkey          info
is_playing key             license
lock      logout         longhelp
loop      move           next
normal    pause          play
playlist  prev           quit
random    rate           repeat
rewind    save_env       sd
search    seek           set
shutdown  slower         snapshot
sort      stats          status
stop      strack         title
title_n   title_p        vcrop
vdeinterlace vdeinterlace_mode vlm
voldown   volume         volup
vratio    vtrack         vzoom
zoom

[2]+  Stopped                  vlc
ismail@localhost [/]$ ps
  PID TTY          TIME CMD
 44606 pts/0    00:00:00 bash
 44781 pts/0    00:00:00 ping
 44791 pts/0    00:00:00 vlc
 44864 pts/0    00:00:00 ps
ismail@localhost [/]$ renice -n 19 -p 44791
44791 (process ID) old priority 0, new priority 19
ismail@localhost [/]$
```